

# Surrogate-Accelerated Conjunction Assessment

Space-filling designs and Gaussian-process emulation for collision probability at catalog scale, with 6-DOF dynamics and EKF/UKF orbit determination

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## Abstract

Screening the tracked-object catalog for collision risk requires evaluating collision probability  $P_c$  for every close approach, and the standard Monte Carlo estimator is prohibitively expensive for the rare-event probabilities involved. We show that it is worse than expensive: across the realistic parameter range,  $P_c$  spans roughly thirteen orders of magnitude, and at  $10^8$  samples brute force returns *zero hits* – no estimate at all – for about three quarters of encounters. We build a Gaussian-process surrogate over the encounter-parameter space, trained at points chosen by parallel-tempering space-filling designs. Two exact invariances of the collision integral reduce its natural six-parameter description to four, a reduction verified numerically to  $\sim 10^{-12}$  and worth an 8 to 14-fold saving in expensive evaluations. The surrogate attains 0.03 orders of magnitude RMSE in  $\log_{10} P_c$  from 64 training runs and answers subsequent queries in 8  $\mu$ s, breaking even against Monte Carlo after 64 conjunctions and running  $\sim 5300$  times cheaper over a 337 789-pair catalog screen. We report two negative results: space-filling designs cannot be substituted for Monte Carlo samples in the rare-event integral itself, and the 2–3 $\times$  advantage optimised designs hold on their own criterion does not translate into a proportional accuracy advantage. The uncertainty above is synthesized, because the public feed carries no covariance; the second half of the report builds the machinery that would measure it instead. We implement a 6-DOF rigid-body propagator, validated against closed-form torque-free precession to  $2 \times 10^{-11}$  rad and against conservation laws to  $2 \times 10^{-13}$ , and compare an extended and an unscented Kalman filter for orbit determination on identical tracking arcs. From TLE-grade initial uncertainty the EKF becomes *confidently wrong* – averaged NEES 1019 against an acceptance band of 5.1 to 7.0 – and we locate the failure rather than reporting it: linear covariance propagation across a tracking gap underestimates the variance of the most tightly known direction by a factor of 400, while per-axis uncertainties remain correct to 1 %. A six-parameter campaign over a space-filling design maps where each filter stays consistent: the EKF at 28 % of design points, the UKF at 72 %.

## 1 Introduction

Two tracked objects pass close to one another. The probability that they collide is not a geometric question, because neither position is known precisely. Orbit uncertainty derived from two-line element sets is of order 1 km, while the objects themselves are of order 1 m and the reported miss distances are hundreds of metres. The uncertainty is three orders of magnitude larger than the objects and larger than the separation being resolved, so only a probability is meaningful.

We measured this gap directly. Rebuilding 37 published conjunctions by propagating both objects from public element sets to the time of closest approach gave separations differing from the reported miss distance by 0.6 km to 2.0 km, against reported values of 0.08 km to 0.43 km (Section 2). Every rebuilt separation was *larger* than reported, which is the expected consequence of adding isotropic error to a near-zero displacement: distance is a norm, so random perturbation almost always increases it.

Catalog-scale screening compounds the problem. Megaconstellation growth is increasing the number of screened pairs faster than compute budgets grow, and the standard Monte Carlo estimator scales as  $1/\sqrt{N}$ , so a tenfold accuracy improvement costs a hundredfold in samples.

## Contributions.

1. A measurement of *where brute force fails outright* rather than merely slowly (Section 3, Figure 1).
2. An exact  $6 \rightarrow 4$  reduction of the encounter-parameter space, verified numerically rather than asserted (Section 5).
3. A Gaussian-process surrogate trained at parallel-tempering space-filling design points, benchmarked against random Latin hypercubes and uniform sampling (Section 6).
4. A triage classifier trained on a catalog screen of 337 789 conjunctions, evaluated at full recall against the baseline an analyst would actually use (Section 7).
5. Two negative results (Sections 4 and 6) that constrain how this family of methods should be applied.
6. A 6-DOF rigid-body propagator validated against closed-form torque-free precession and against conservation laws, with the quaternion norm constraint handled explicitly rather than hidden (Section 8).
7. An EKF/UKF comparison in which the EKF’s loss of consistency is *located* – a collapse of one covariance direction during propagation – and a demonstration that per-axis  $3\sigma$  bounds fail to detect it (Section 9).
8. A six-parameter campaign mapping where each filter remains consistent, executed reproducibly in parallel and in containers (Section 10).

## 2 Propagation and uncertainty

Element sets are retrieved from CelesTrak and Space-Track and propagated with SGP4 [1, 2]. TLE elements are Brouwer mean elements defined by the SGP4 theory itself; propagating them with a general-purpose orbit integrator introduces kilometre-level error immediately, so the element set and the propagator are used strictly as a matched pair. Propagation was validated against the International Space Station: mean altitude 419.4 km and period 92.96 min from the elements, against 92.91 min measured independently by counting minima of geocentric radius over three days of propagation.

The public conjunction feed carries no covariance data. It reports time of closest approach, miss distance, object identifiers, radar cross-section class and the originator’s own probability,

but no state vectors and no covariance matrices. Uncertainty must therefore be modelled, and we state it as an assumption rather than presenting it as a data product.

Each object is assigned a diagonal covariance in its own radial / in-track / cross-track (RTN) frame,

$$C_{\text{RTN}} = \text{diag}(\sigma_R^2, (k_T \sigma_R)^2, (k_N \sigma_R)^2), \quad (1)$$

with anisotropy ratios fixed at  $k_T = 10$ ,  $k_N = 1.5$  and only the scale  $\sigma_R$  fitted. In-track error dominates because a small period error integrates into a growing along-track lag. Fixing the ratios is deliberate: a full  $3 \times 3$  covariance per object has twelve free parameters, and the data provides one scalar observable per event. Fitting the scale alone to the rebuilt-versus-reported miss spread gives

$$\sigma_R = 95 \text{ m}, \quad \sigma_T = 953 \text{ m}, \quad \sigma_N = 143 \text{ m},$$

magnitudes consistent with published TLE accuracy and obtained from our own measurements rather than assumed. The two objects occupy different orbital planes, so each covariance is rotated into a common inertial frame before the relative covariance  $C = C_1 + C_2$  is formed.

**What this model does not do.** Per-event correlation between predicted and observed error is  $-0.53$ . The model reproduces the aggregate *scale* of TLE uncertainty but not the per-event ranking; it does not identify which conjunctions are worst. A single scalar observable per event is insufficient to constrain that, and we do not claim otherwise.

### 3 Collision probability and the cost of brute force

At the relative speeds typical of low Earth orbit ( $10 \text{ km s}^{-1}$  to  $15 \text{ km s}^{-1}$ ) an encounter lasts milliseconds and relative motion is effectively rectilinear. The three-dimensional problem then collapses onto the two-dimensional encounter plane perpendicular to the relative velocity [3, 4], and

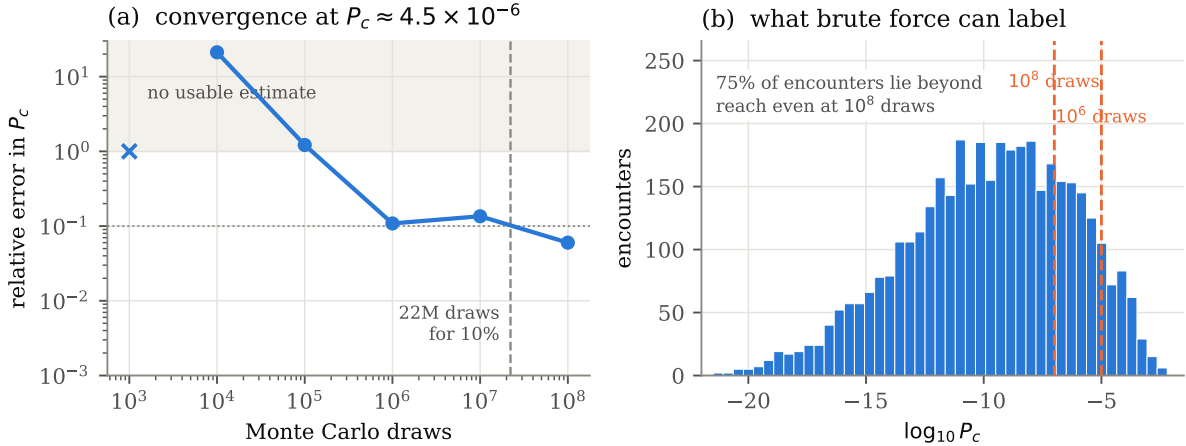
$$P_c = \int_{\|x\| \leq R} \mathcal{N}(x; \mu, C) \, dx, \quad (2)$$

where  $\mu$  and  $C$  are the projected relative miss vector and covariance and  $R$  is the combined hard-body radius.

We implement three independent estimators of (2): a batched Monte Carlo sampler, polar quadrature with resolution adapted to the covariance, and a small-disk closed form valid when  $R \ll \sigma$  [5]. Monte Carlo and quadrature agree within Monte Carlo error bars on every real conjunction tested (maximum  $|z| = 1.4$ ), which cross-validates the encounter geometry and the sampler against each other.

Figure 1 states the problem quantitatively. At  $P_c \approx 4.5 \times 10^{-6}$ , 100 000 samples return *zero hits*: not a poor estimate but no estimate. Reaching 10 % relative error requires  $\sim 22\,000\,000$  samples and 1 % requires  $\sim 2.2 \times 10^9$ , per conjunction. Across the parameter box  $P_c$  spans about thirteen orders of magnitude, and a substantial fraction of encounters sit below  $10^{-12}$ , where no achievable sample budget yields a single hit.

This has a direct methodological consequence: **training labels for the surrogate cannot come from Monte Carlo**, because Monte Carlo cannot label most of the space. We use the quadrature, which is exact in this Gaussian setting and agrees with Monte Carlo wherever Monte Carlo works at all. The sampler stands in for the general case – non-Gaussian uncertainty, long-duration encounters, non-spherical bodies – where no closed form exists.



**Figure 1:** Brute force does not merely converge slowly; over most of the parameter range it returns nothing. **(a)** Monte Carlo convergence at a representative  $P_c \approx 4.5 \times 10^{-6}$ : below  $10^5$  samples the estimator produces zero hits, and 22 000 000 samples are needed for 10% relative error. **(b)** Distribution of  $P_c$  over the encounter-parameter box, with the reach of two sample budgets marked at ten expected hits. Roughly three quarters of encounters lie beyond reach even at  $10^8$  samples.

## 4 A negative result: designs cannot replace samples

The intuitive reading of “use a space-filling design instead of Monte Carlo” is to replace the random samples in (2) with a well-spread deterministic design. This does not work, and the reason is worth stating because the idea is natural.

Equation (2) is the integral of an indicator function over a microscopic region. A 256-point space-filling design returns zero hits just as  $10^5$  random samples do, and in fact more reliably, because such a design deliberately spreads points *away* from any small region. Quasi-Monte Carlo does not rescue the construction either: its convergence advantage requires a smooth integrand, and an indicator over a tiny region is the opposite of smooth.

The construction that does work places the design in the **encounter-parameter space** rather than the sample space. Each design point receives one expensive evaluation, a surrogate interpolates between them, and the cost is amortised across every subsequent query.

## 5 Dimensional reduction

The integral (2) appears to depend on six parameters:  $\mu$  (two components),  $C$  (three), and  $R$  (one). Only four are independent. The integration domain  $\|x\| \leq R$  is rotation-symmetric, so rotating  $\mu$  and  $C$  together leaves  $P_c$  unchanged; and the integral is covariant under uniform scaling of all lengths, so dividing  $\mu$ ,  $\sqrt{C}$  and  $R$  by a common factor leaves it unchanged. Rotating into the eigenbasis of  $C$  and scaling by  $R$  therefore removes two parameters, leaving  $(\mu_1/R, \mu_2/R, \sigma_1/R, \sigma_2/R)$ .

Because a silently incorrect reduction would invalidate everything downstream, the invariances were verified numerically rather than accepted from the algebra (Table 1).

The saving is substantial. Measured fill distance at  $k = 4$ ,  $n = 64$  is 0.397; matching it in six dimensions requires  $\sim 500$  to 900 points, an 8 to 14-fold increase in expensive evaluations for

**Table 1:** Numerical verification of the reduction over random encounters spanning the full parameter box. Residuals are at the level of quadrature noise: these are exact identities, not approximations.

| Claim                                           | Cases | Max. relative error   |
|-------------------------------------------------|-------|-----------------------|
| $P_c$ invariant under rotation of $\mu$ and $C$ | 300   | $8.3 \times 10^{-13}$ |
| $P_c$ invariant under uniform length scaling    | 300   | $9.8 \times 10^{-13}$ |
| Reduced form reproduces $P_c$                   | 200   | $8.4 \times 10^{-13}$ |
| $P_c$ even in each $\mu$ component (eigenbasis) | 200   | $6.2 \times 10^{-16}$ |
| $6 \rightarrow 4 \rightarrow P_c$ round trip    | 200   | $5.5 \times 10^{-13}$ |

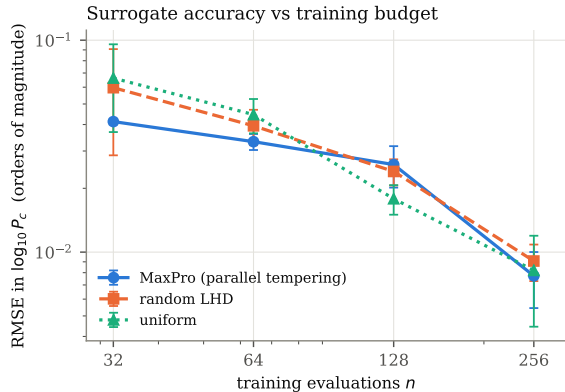
identical answers. The sign symmetry, exact to  $6 \times 10^{-16}$ , permits restricting the domain to  $\mu_1, \mu_2 \geq 0$ , reducing its volume by a further factor of four.

The identity holds *given the model*: a circular hard body, Gaussian uncertainty, and the short-term encounter approximation. Slow encounters, below about  $1 \text{ km s}^{-1}$  relative speed, fall outside it and are excluded here.

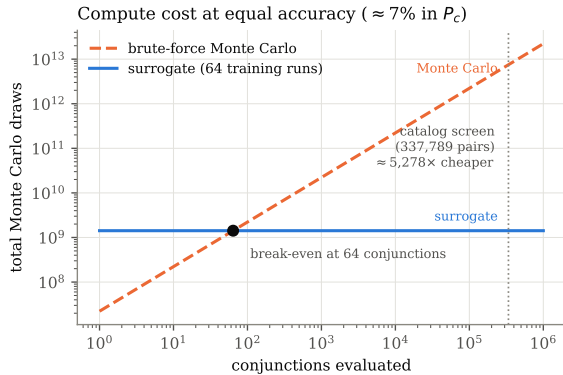
## 6 Surrogate

Design points are generated by parallel-tempering [9] optimisation of the MaxPro criterion [8], reusing the C++ implementation developed in our prior work on space-filling designs [11] and wrapped for Python here. Designs are static artefacts: the optimiser runs once per  $(n, k)$  and costs 0.2s to 1.4s. On its own criterion the optimised design beats the best of twenty random Latin hypercubes [6] by 2.1–3.1 $\times$ ; the Morris–Mitchell  $\phi_p$  criterion [7] is reported alongside it in the project log.

The surrogate is a Gaussian process [10] with an anisotropic squared-exponential kernel, hyperparameters obtained by maximising the log marginal likelihood. Targets are  $\log_{10} P_c$ : with thirteen orders of magnitude of dynamic range, a GP fitted to  $P_c$  itself would be dominated entirely by the few largest values.



**Figure 2:** Surrogate accuracy against training budget, five replicates per design type; error bars are the standard deviation across replicates. All three converge by  $n = 256$ .



**Figure 3:** Total compute to evaluate  $Q$  conjunctions at matched accuracy ( $\approx 7\%$  in  $P_c$ ). The surrogate’s cost is flat after training; break-even is 64 conjunctions.

At  $n = 64$  the surrogate attains an RMSE of about 0.03 orders of magnitude in  $\log_{10} P_c$  – roughly 7% in  $P_c$  – and answers subsequent queries in 8.3  $\mu$ s. Figure 3 converts this into the quantity that matters at catalog scale. Charging the surrogate its full training cost and Monte Carlo its per-query cost at matched accuracy, the two break even at 64 conjunctions; over the 337 789-pair screen of Section 7 the surrogate is  $\sim 5300$  times cheaper.

**A second negative result.** The  $2\text{--}3\times$  advantage the optimised designs hold on the MaxPro criterion *does not* translate into a proportional accuracy advantage. Mean RMSE improves by only  $1.1\text{--}1.8\times$  over random Latin hypercubes and uniform sampling at  $n \leq 128$ , and by  $n = 256$  the three are indistinguishable (Figure 2). The target function is smooth, so once the box is adequately covered the design ceases to matter. Better space-filling by the criterion is not the same thing as better surrogate accuracy.

Where the optimised designs do show a consistent advantage is **variance**. Across replicates their RMSE spread is several times tighter than that of random designs. With a one-shot expensive evaluation budget a random design is a lottery – some draws are as good as an optimised one, others substantially worse – and reproducibility is worth as much to a practitioner as the mean improvement.

## 7 Risk triage

The public conjunction feed contains no negative examples: every record is flagged reportable with  $P_c > 10^{-4}$ . A classifier trained on it learns to answer “yes”. We therefore manufactured negatives by screening debris catalogs directly – dense, co-orbital populations that generate many close approaches per object.

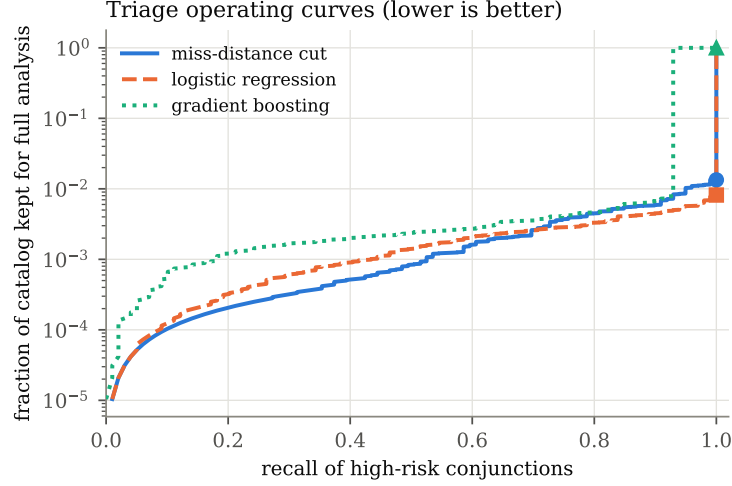
Screening uses a coarse propagation grid, per-pair local minima of separation detected on a rolling three-step window, an analytic linear-motion filter, and parabolic refinement of the survivors. The coarse gate must be wide: at  $16\text{ km s}^{-1}$  a 60 s grid steps 960 km, so a pair passing within 1 km may never appear closer than  $\sim 450$  km at any sampled instant. The gate is consequently set to 530 km, which leaves nearly every pair a candidate; the analytic filter, which locates the rectilinear closest approach in closed form, then removes 91 % of candidates with identical output.

The resulting dataset holds 337 789 conjunctions from three debris families over seven days, with 352 positives (0.104 %) at  $\log_{10} P_c > -10$ . No conjunction in this population reached the operational  $10^{-4}$  threshold, so the label threshold is a stated choice; the ordering of methods is unchanged at  $10^{-8}$  and  $10^{-12}$ .

Features are restricted to quantities available *before* any  $P_c$  computation – miss distance, relative speed, altitude, relative inclination, element-set age, period difference, eccentricity and drag term. Including  $P_c$  or anything derived from it would make the task circular.

Accuracy and AUC are near-meaningless at a 0.1 % positive rate; a constant negative prediction scores 99.9 %. Because missing a high-risk conjunction is unacceptable while passing a harmless one costs only compute, models are compared by how much of the catalog they discard *at full recall*. The baseline is a miss-distance cut, which is what an analyst does without a model, and splits are grouped by day or by object rather than at random, since conjunctions in the same window share objects and geometry.

Logistic regression passes 0.82 % to 0.92 % of conjunctions against 1.15 % to 1.41 % for the baseline, and on the object split it retains full recall where the baseline misses one. Miss distance



**Figure 4:** Triage operating curves on held-out data. Below 70 % recall a miss-distance cut is the better ranker; the learned model wins at the high-recall end, which is the operationally relevant one. Gradient boosting reaches full recall only by retaining the entire catalog.

**Table 2:** Triage performance. “Kept” is the fraction of conjunctions passed on for expensive analysis; lower is better at equal recall. Thresholds are set from out-of-fold predictions on training data and applied unchanged.

| Split  | Model                        | Kept    | Recall  | Missed |
|--------|------------------------------|---------|---------|--------|
| Day    | Miss-distance cut (baseline) | 1.41 %  | 100.0 % | 0      |
|        | Logistic regression          | 0.92 %  | 100.0 % | 0      |
|        | Gradient boosting            | 100.0 % | 100.0 % | 0      |
| Object | Miss-distance cut (baseline) | 1.15 %  | 97.2 %  | 1      |
|        | Logistic regression          | 0.82 %  | 100.0 % | 0      |
|        | Gradient boosting            | 100.0 % | 100.0 % | 0      |

alone has a rank correlation of only +0.60 with  $\log_{10} P_c$ : the projected covariance depends on encounter geometry that a single distance does not capture.

Gradient boosting fails outright, retaining the entire catalog under every configuration. This is not a threshold artefact – its oracle operating point is also  $\approx 100\%$ , meaning the lowest-scoring true positive ranks below almost the whole test set. At a 0.1% positive rate the flexible model fragments a space in which the positives occupy a thin tail and cannot rank them, while a model whose score is monotone in roughly the right direction can. Model capacity actively hurts.

## 8 A 6-DOF rigid-body propagator

The covariance used in Section 2 is synthesized, which is the most serious limitation of everything above: the public feed carries no covariance, so an error model had to be assumed and calibrated against one scalar observable per event. An orbit-determination filter produces a covariance *from measurements* instead. Building one requires truth trajectories from dynamics under our own control, which is what this section supplies.

**State and equations of motion.** The state is thirteen elements: position  $\mathbf{r}$  and velocity  $\mathbf{v}$  of the centre of mass (km, km s<sup>-1</sup>, J2000 ECI), the attitude quaternion  $q$  (scalar-first, Hamilton convention, rotating body axes into the inertial frame), and the body angular velocity  $\boldsymbol{\omega}$  (rad s<sup>-1</sup>, body axes). Translation is Newton’s law per unit mass; rotation is quaternion kinematics with Euler’s equations [12]:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_{J_2}, \quad \dot{q} = \frac{1}{2}q \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}, \quad I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times I\boldsymbol{\omega}. \quad (3)$$

Force and torque models are supplied through a protocol; the ones implemented are two-body gravity, the  $J_2$  oblateness perturbation, and the gravity-gradient torque  $\boldsymbol{\tau}_{gg} = 3\mu r^{-3}\hat{\mathbf{u}} \times I\hat{\mathbf{u}}$  with  $\hat{\mathbf{u}}$  the unit radius vector in body axes. Translation and rotation couple only through those models. Quaternion-to-matrix conversion uses Shepperd’s method [13], which stays accurate near 180° where the naive trace formula loses all precision.

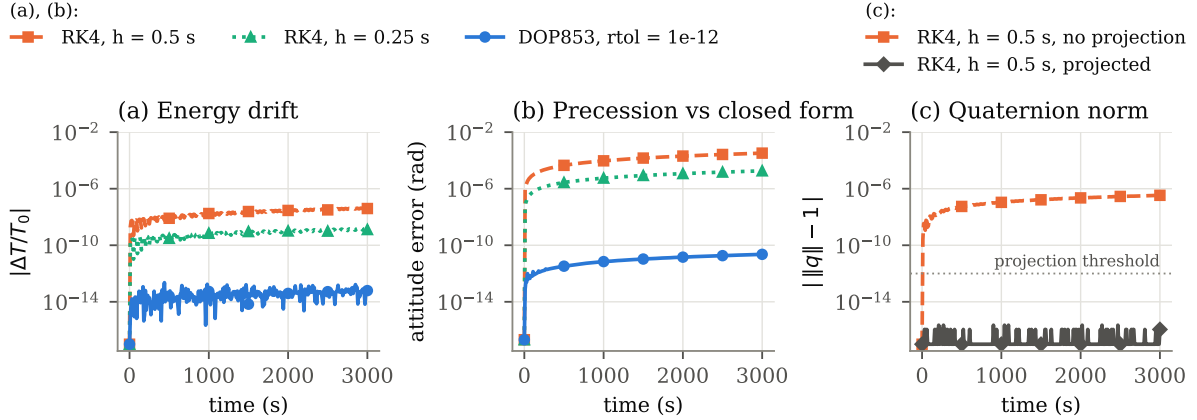
**Integration and the norm constraint.** A fixed-step RK4 and an adaptive DOP853 [14] sit behind one interface, so the two can be compared on identical problems. The exact flow preserves  $\|q\| = 1$  because  $q \cdot \dot{q} = 0$ , but a discrete integrator does not, and the drift is handled explicitly rather than hidden inside the equations: each integrator accepts a constraint, projects the quaternion back to unit norm when the violation exceeds a threshold, and reports how often it did so. DOP853 offers no hook between steps, so the violation is detected as a terminal event, the state projected, and the integration restarted.

Table 3 lists the validation; Figure 5 shows the error growth behind three of its rows. Two entries deserve comment. The quaternion norm drifts to  $3.4 \times 10^{-7}$  unprojected and holds at  $2.2 \times 10^{-16}$  with projection, which is the constraint machinery earning its place. The  $J_2$  nodal rate agrees to 0.42% rather than exactly, because the analytic rate is expressed in mean elements while the simulation is initialised with osculating ones – a discrepancy of order  $J_2$  itself, which is the size observed.

At matched accuracy the adaptive integrator is substantially cheaper: over the same 3000 s tumble DOP853 uses 27 200 right-hand-side evaluations for an energy drift of  $2.3 \times 10^{-13}$ , against 48 000 for RK4 at  $1.5 \times 10^{-9}$ .

**Table 3:** Validation of the propagator. Every reference is an identity or a closed-form solution, not a stored output; tolerances that are not exact derive from a stated source of error.

| Check                            | Reference                                                                 | Result                                               |
|----------------------------------|---------------------------------------------------------------------------|------------------------------------------------------|
| Torque-free energy drift, 3000 s | $T = \frac{1}{2}\boldsymbol{\omega} \cdot I\boldsymbol{\omega}$ conserved | $2.3 \times 10^{-13}$                                |
| Torque-free momentum drift       | $\mathbf{H}$ conserved as a vector                                        | $5.7 \times 10^{-12}$                                |
| RK4 step halving                 | fourth-order convergence                                                  | $> 16\times$ reduction                               |
| Axisymmetric precession          | closed form, $\dot{\phi} = \ H\ /I_t$                                     | $2.3 \times 10^{-11}$ rad                            |
| Quaternion norm, RK4 $h = 0.5$ s | unit norm                                                                 | $3.4 \times 10^{-7} \rightarrow 2.2 \times 10^{-16}$ |
| Two-body orbit after one period  | closes on itself                                                          | $< 1 \times 10^{-6}$ km                              |
| $J_2$ nodal regression           | $-\frac{3}{2}nJ_2(R/a)^2 \cos i$                                          | within 0.42 %                                        |
| Gravity-gradient torque          | direct sum over point masses                                              | $1 \times 10^{-5}$                                   |
| Small-angle pitch libration      | $\omega = n\sqrt{3(I_2 - I_1)/I_3}$                                       | within 0.1 %                                         |



**Figure 5:** Propagator validation over a 3000s torque-free tumble. (a) rotational energy drift; (b) attitude error against the closed-form axisymmetric precession solution; (c) quaternion norm drift with and without projection, with the projection threshold marked. Values are clamped at  $1 \times 10^{-17}$  so that exact zeros remain plottable on log axes.

## 9 Orbit determination: EKF versus UKF

The filter state is  $x = [\mathbf{r}, \mathbf{v}]$  in J2000 ECI, propagated with the same force models as the truth, so the comparison isolates the filters rather than a modelling difference. The EKF propagates its covariance with the state transition matrix obtained by integrating the variational equations  $\dot{\Phi} = A\Phi$  alongside the state [15], where the gradient  $\partial \mathbf{a} / \partial \mathbf{r}$  is taken by central differences so that a new force model needs no hand-derived Jacobian, and uses the Joseph form of the covariance update, which stays symmetric positive semi-definite under round-off [18]. The UKF [16] pushes thirteen scaled sigma points [17] through the full nonlinear dynamics and measurement model. Its parameters are  $\alpha = 1$ ,  $\beta = 2$ ,  $\kappa = 0$ : all weights are then non-negative and the predicted covariance is positive semi-definite by construction, whereas the frequently quoted  $\alpha = 10^{-3}$  gives a centre weight of about  $-10^6$  in six dimensions.

Measurements are range and range-rate (10 m,  $1 \text{ cm s}^{-1}$ ) from three widely separated ground stations with a  $10^\circ$  elevation mask, taken every 60 s while an object is in view. Station positions are WGS-84 geodetic coordinates rotated by Greenwich mean sidereal time. Sensors are added by implementing a measurement protocol; neither filter changes.

Consistency is judged by chi-square tests rather than by error magnitude alone [18]. The normalised estimation error squared,  $\epsilon = e^\top P^{-1}e$ , is  $\chi^2$  with six degrees of freedom for a consistent filter; averaged over  $N$  independent runs it must fall inside a two-sided interval derived from  $\chi^2(6N)/N$ . An average above the interval means the filter’s covariance is too small for its own errors: overconfident, and therefore inclined to discount the measurements that would correct it.

**Table 4:** Fifty Monte Carlo runs per case, two orbital revolutions, three stations. Only the initial uncertainty differs between cases. The 95 % acceptance band for averaged NEES is 5.1 to 7.0.

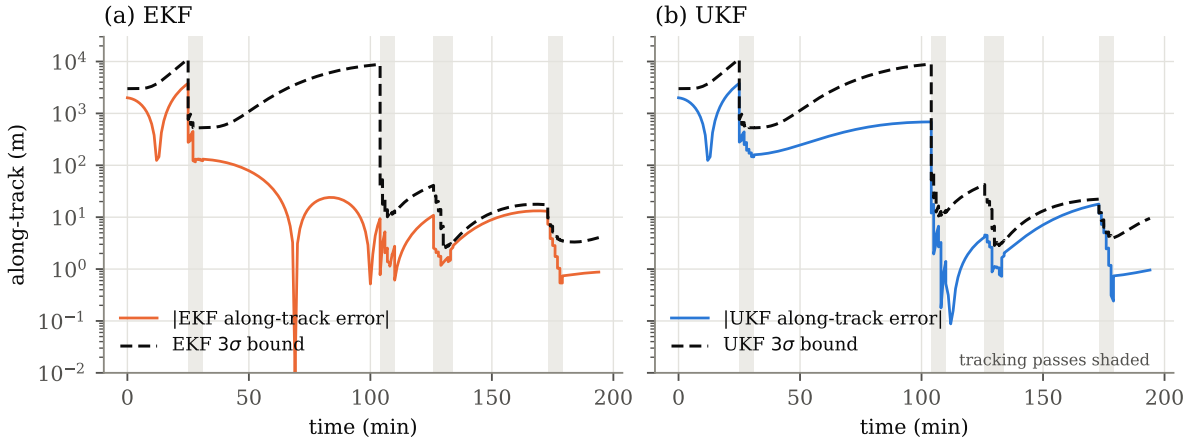
| Initial uncertainty (per axis)          | Filter | Final NEES | Final position RMSE |
|-----------------------------------------|--------|------------|---------------------|
| 10 m / $1 \text{ cm s}^{-1}$            | EKF    | 6.6        | 3.0 m               |
|                                         | UKF    | 6.6        | 3.0 m               |
| 1 km / $1 \text{ m s}^{-1}$ (TLE grade) | EKF    | 1019       | 22.8 m              |
|                                         | UKF    | 8.8        | 5.8 m               |

With precise initialisation the two filters are indistinguishable and both consistent (Table 4). Given initial uncertainty of 1 km – roughly what a TLE supplies, and therefore the realistic case for the population screened in Section 7 – the EKF’s averaged NEES reaches 1019 and its final error is four times the UKF’s.

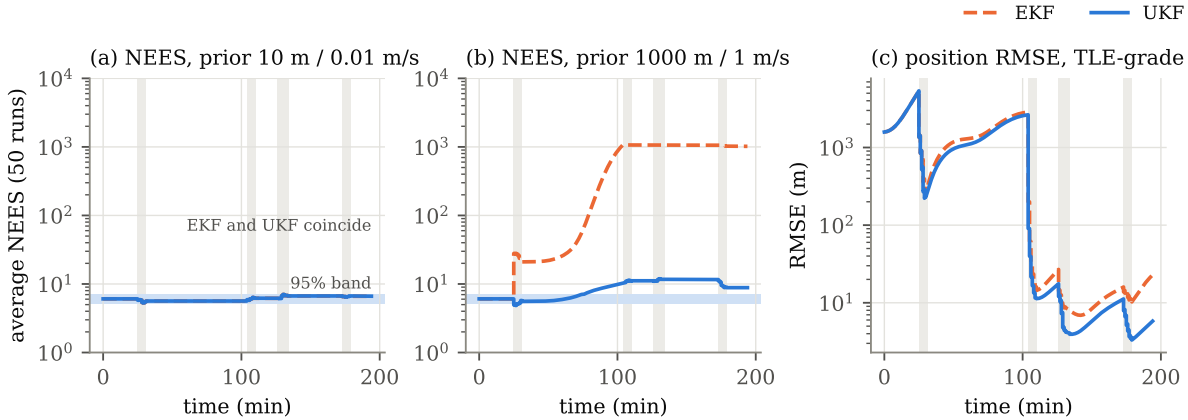
**Locating the failure.** Two mechanisms contribute, and the smaller one is the more obvious. At the first measurement update the prior is 1 km to 4 km wide, and the second-order term the EKF discards has a standard deviation 3.2 times the range-rate noise (against 0.36 for range); NEES rises from 5 to about 20. That is not what produces 1019. The dominant effect is in the *prediction* across the 73 min gap between the first and second tracking passes. A pass leaves the covariance extremely anisotropic, its eigenvalues spanning some ten orders of magnitude. Second-order dynamics transfer variance from the wide directions into the narrow ones; linear propagation  $\Phi P \Phi^\top$  cannot create it. Pushing a 4000-point cloud through the full nonlinear dynamics gives a smallest eigenvalue of  $1.5 \times 10^{-12}$ , and the UKF predicts  $1.7 \times 10^{-12}$ , while the

EKF predicts  $3.6 \times 10^{-15}$  – low by a factor of 400. The cloud’s NEES under the EKF covariance is 1104; under the UKF’s it is 5.3.

The diagnostic value of this is methodological. The mean and the per-axis radial, in-track and cross-track standard deviations agree with the true distribution to 1% for *both* filters. The EKF is wrong only in the one direction it believes it knows best – which is precisely the direction NEES weights most heavily, and the direction whose small variance causes the next measurement to be over-trusted. Consequently the customary per-axis  $3\sigma$  plot looks entirely healthy (Figure 6) while the averaged NEES is  $150\times$  above its acceptance band (Figure 7). **Plotting error against per-axis  $3\sigma$  bounds is not a consistency test.**



**Figure 6:** One run from the TLE-grade case: along-track error against the filter’s own  $3\sigma$  bound, tracking passes shaded. Neither panel shows a violation, and on this evidence both filters look sound – Figure 7 shows that one of them is not.



**Figure 7:** Averaged NEES over fifty runs with the 95% acceptance band shaded, for (a) the precise prior, where the filters coincide, and (b) the TLE-grade prior, where the EKF departs during the gap after the first pass. (c) position RMSE for the TLE-grade case.

The UKF is not vindicated unconditionally. Its averaged NEES in the TLE-grade case settles at 8.8, above the band’s upper limit of 7.0: second-order moment matching is itself an

approximation. The distinction is one of kind rather than degree – a small factor against three orders of magnitude.

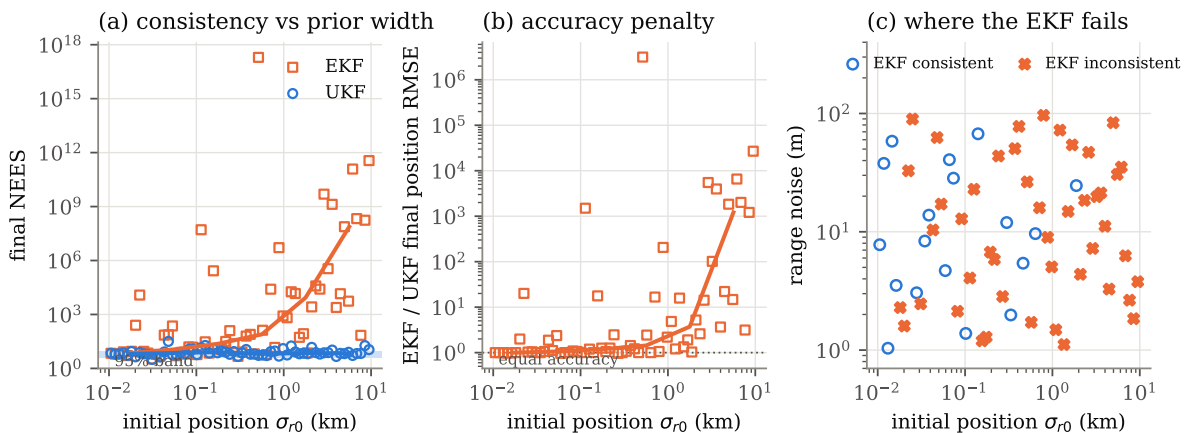
## 10 Where each filter holds: a six-parameter campaign

Two points in parameter space do not establish a boundary. We therefore swept six parameters – initial position and velocity uncertainty, range and range-rate noise, measurement cadence, and elevation mask – over a MaxPro space-filling design [11], the same machinery used for the surrogate in Section 6, and ran a Monte Carlo with both filters at each of 64 design points.

**Table 5:** Campaign over 64 design points, eight trials per point. Medians, because NEES spans orders of magnitude across the design and a mean would report only its worst points.

| Filter | Points consistent | Median NEES | 90th pct. NEES    | Median RMSE |
|--------|-------------------|-------------|-------------------|-------------|
| EKF    | 28 %              | 70.2        | $1.4 \times 10^8$ | 10.4 m      |
| UKF    | 72 %              | 7.0         | 12.5              | 3.0 m       |

The sweep converts the two-point result into a trend (Figure 8). Binned by initial position uncertainty over decades from 10 m to 10 km, the EKF is consistent at 50 %, 38 %, 19 % and 6 % of points, while the UKF stays near 75 % throughout. The accuracy penalty has the same shape: the median ratio of final position error, EKF over UKF, is 1.30, but the 90th percentile is  $1.7 \times 10^3$ , because beyond roughly 2 km of prior uncertainty some EKF runs diverge outright instead of degrading. Over the whole design the UKF’s worst case is NEES 32 and 185 m; the EKF’s is  $2 \times 10^{17}$  and 2570 km. Panel (c) of Figure 8 shows the failures organised by prior width rather than by sensor noise, which is the practically useful statement: it is the quality of the initial orbit, not the quality of the radar, that decides whether the cheaper filter is safe.



**Figure 8:** Campaign results. (a) final NEES against initial position uncertainty, with the acceptance band shaded and binned medians overlaid; (b) ratio of final position RMSE, EKF over UKF; (c) design points at which the EKF is and is not consistent, against prior width and range noise.

**Reproducibility as an engineering property.** Each design point derives its random stream from the pair (campaign seed, point index), so results do not depend on the number of worker

processes, the number of shards, or the order in which they finish. The campaign runs locally across cores and, unchanged, as an array job on a batch service, with the shard index read from the environment. A two-shard, two-worker run inside the container is bit-identical to a single-worker unsharded run; across platforms the agreement is about  $10^{-5}$  relative, because the two link different BLAS implementations. Raw rows are persisted as Parquet beside the configuration that produced them, its hash, and the library versions.

## 11 Limitations

1. **Absolute probabilities are not validated.** Our  $P_c$  values run 10–100× below the originator’s published figures. The dominant suspect is the hard-body radius, since  $P_c \propto R^2$  and radar-cross-section-derived radii of 1 m to 4 m are likely below operational values. Agreement with published probabilities is not claimed as validation.
2. **Uncertainty is synthesized, not measured,** and reproduces the aggregate scale of TLE error rather than its per-event variation.
3. **The triage population is debris,** which is small, uncontrolled and co-orbital; operational payload conjunctions differ. Test sets contain 36–99 positives, so differences of a few tenths of a percent should not be over-read.
4. **Seven-day propagation from a single epoch** exceeds the 2 to 5-day range over which the covariance model was calibrated.
5. **The dimensional reduction is model-conditional** and does not extend to non-Gaussian uncertainty, non-circular bodies, or long-duration encounters.
6. **The two halves of this report are not yet coupled.** SGP4 outputs TEME while the 6-DOF dynamics work in J2000 ECI, and the conversion between them is deliberately not implemented, so no state crosses from one to the other. The frames differ by the equation of the equinoxes – hundreds of metres to kilometres in low Earth orbit – which is precisely the scale of the uncertainty under study, so mixing them silently would corrupt both. Feeding a filter-derived covariance into  $P_c$ , which is what would replace the synthesized model, requires that conversion first.
7. **The Earth-orientation model is simplified.** Rotation is taken about the J2000  $z$ -axis at the sidereal rate with Greenwich mean sidereal time, neglecting precession, nutation and polar motion, and UT1 is identified with UTC. This is self-consistent within a simulation, where truth and filter share it, but not adequate for processing real tracking data.
8. **The estimation results contain no dynamic mismodelling.** Truth and filter use identical force models and the process noise is zero, which isolates the effect of the filters’ own approximations. Real orbit determination also contends with unmodelled drag and solar radiation pressure, which this comparison does not exercise.
9. **The UKF is not unconditionally consistent** either: it sits somewhat above the acceptance band in the TLE-grade case and misses it at about a quarter of the campaign’s design points, by small factors rather than orders of magnitude.

## 12 Reproducibility

All results are produced by scripts in the accompanying repository. Every important quantity is computed two independent ways: Monte Carlo against quadrature, the C++ design criterion against a `numpy` reimplementaion (agreeing to six decimal places), the orbital period from elements against a period measured by counting radius minima, and the state transition matrix against finite differences and against its own symplecticity. The test suite comprises 389 offline checks covering 96 % of package statements, and asserts physical identities – conservation laws, closed-form solutions, exact invariances, chi-square consistency bands – in preference to stored values. Continuous integration runs the suite, a linter enforcing `numpy`-style docstrings, a strict type check, and a build of the campaign container on every push.

**Table 6:** Commands reproducing each result.

| Command                                       | Produces                        |
|-----------------------------------------------|---------------------------------|
| <code>python scripts/validate_iss.py</code>   | Propagation validation (Sec. 2) |
| <code>python scripts/calibrate.py</code>      | Covariance calibration (Sec. 2) |
| <code>python scripts/run_montecarlo.py</code> | Cost baseline (Sec. 3)          |
| <code>pytest tests/test_paramspace.py</code>  | Table 1                         |
| <code>python scripts/run_surrogate.py</code>  | Figure 2                        |
| <code>python scripts/build_dataset.py</code>  | Triage dataset (Sec. 7)         |
| <code>python scripts/run_triage.py</code>     | Table 2                         |
| <code>python scripts/validate_6dof.py</code>  | Table 3, Figure 5               |
| <code>python scripts/run_estimation.py</code> | Table 4, Figures 6–7            |
| <code>python scripts/run_campaign.py</code>   | Table 5, Figure 8               |
| <code>python scripts/make_figures.py</code>   | Figures 1–4                     |

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